Bring KANs to Tracker: A Nonlinear Fusion Strategy Over Frequency Decoupling for Temporal Hyperspectral Object Tracking

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***Abstract*—Hyperspectral (HS) cameras have great potential in extracting spectral, textual, and temporal information from objects. Many existing works leverage HS data for object tracking, as it provides unique spectral features that can help address challenges like background clutter (BC) or camouflage. However, most of these methods overlook the rich temporal information available in video sequences, and many spectral–visual fusion approaches fail to extract contextual information from a global perspective, causing an insufficient understanding of an entire object by the model. To address the above issues, a spectral–temporal tracking Transformer based on a frequency-domain fusion strategy (S3T-FFS) is proposed. First, a spectral–temporal token is proposed to capture an object's spectral information that remains unchanged in video clips, providing additional tracking cues. Second, to extract spectral semantic information, we propose a frequency-domain fusion strategy (FFS), including a frequency attention network (FAN) and a Kolmogorov–Arnold network-based convolutional unit (CuKAN), to provide spectral information for the tracking model. Specifically, FAN is designed for the simultaneous extraction and fusion of spectral features. This synchronous modeling approach decouples low-frequency and high-frequency information, adjusting their balance through frequency-domain prior knowledge and self-adaptive weights. This allows the fusion network to focus more on global information. To further extract global patterns from the low-frequency features and improve the network's interpretability, we introduce CuKAN to extract nonlinear relationships within the decoupled frequency components, and its learnable activation function helps the model learn global patterns, thus avoiding the local overfitting in convolutional neural networks. Extensive experiments on multiple large-scale datasets illustrate the effectiveness of our proposed methods.**

***Index Terms*—Deep learning, frequency attention, hyperspectral object tracking (HOT), Kolmogorov–Arnold network.**

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I. Introduction

HYPERSPECTRAL object tracking (HOT) is a promising technology for detecting objects using hyperspectral (HS) imaging methods. Unlike traditional RGB images, hyperspectral images (HSIs) capture spectral information across dozens to hundreds of consecutive bands for each pixel, giving them unique advantages in object detection and recognition [1], [2]. By capturing rich spectral information, HS imaging distinguishes objects with similar colors but distinct spectral characteristics more accurately. This technology has broad applications in fields such as remote sensing [3], [4] and environmental monitoring [5]. For example, detecting camouflaged objects or monitoring crop health [6] in agriculture can be achieved by analyzing the spectral characteristics of the objects.

Currently, researchers are exploring deep learning methods for HS images. Existing methods typically use a multistage pipeline, as shown in Fig. 1, and can be divided into two categories. The first is based on band regrouping [7], [8], where spectral bands are reorganized into multiple three-channel false-color images to expand the training data. Deep learning networks are then built for training, with regression and classification heads working together to infer the object's position. The other category involves spectral–visual feature fusion methods [9], [10], which use the spectral information from HS images alongside the visual information from RGB images for feature extraction. This enhances feature robustness and creates more discriminative object representations, outperforming the band regrouping method. However, both methods primarily model relationships between independent frames, overlooking crucial temporal cues. Extracting temporal features from continuously moving objects across consecutive frames can better handle situations where spectral–visual information is disrupted, such as during occlusion (OCC). Furthermore, the existing modality fusion methods face difficulties in capturing contextual information from a global perspective, making it challenging to generate effective cross-modality feature representations.

To address the aforementioned issues, we propose S3T-FFS, a spectral–temporal tracking framework designed for time-series modeling and frequency-domain attentive fusion using a KAN-based architecture. Temporal information has been shown to be an effective structure in visual object tracking [11], [12], as it provides video-level contextual details.

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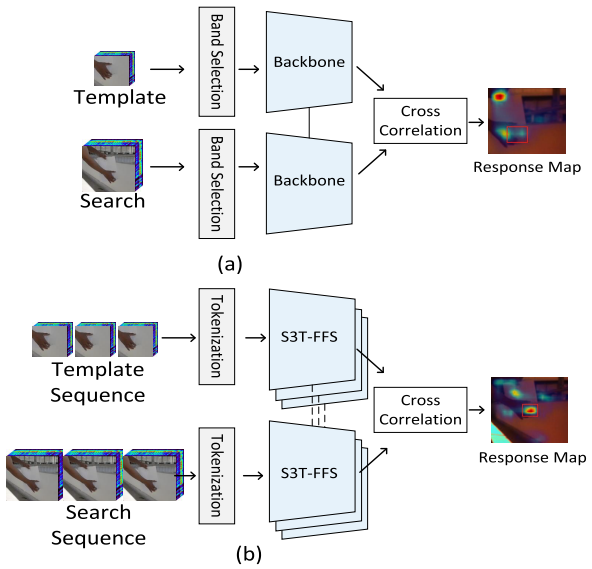


Fig. 1. Comparison of tracking pipeline. The visualization of heatmaps on the Card sequence of the HOTC-2020 dataset. The red color represents high response, while the blue color represents low response. The tracked object is marked with a red box. (a) Existing pipeline. (b) Our pipeline.

However, it has not yet been fully explored in the context of HOT. Our core idea is that while an object's appearance in an HS video may be affected by disturbances such as rotation, deformation (DEF), or shaking, its spectral information remains largely invariant. Preserving this spectral invariance across temporal associations in frames is therefore crucial. In S3T-FFS, we shift from single-frame sampling to multiframe, video-level sampling and introduce a spectral–temporal token to link the spectral information between the template and the search frames. The spectral token aggregates information from all other tokens through the self-attention mechanism of the Transformer encoder, enabling global feature aggregation. This approach implicitly captures the relationships between multiple templates and temporal tokens, as well as between multiple queries and temporal tokens. Compared to previous methods [11], [12], [13], the spectral–temporal token requires only a small number of training parameters and does not rely on any online template update strategy.

Tracking objects is often accompanied by adjacent appendages, such as hands holding objects, intersecting vehicles, or obstructing trees. These appendages interfere with spectral–temporal learning and disrupt the temporal consistency of the tracked objects. Additionally, existing frameworks tend to over-rely on fine-grained spatial or spectral features, leading to local feature overfitting. As a result, during inference, the network may become biased toward similar appendages, hindering its ability to learn robust global patterns. For instance, as shown in Fig. 1(a), the peak value of the response map has shifted to a hand adjacent to the object. To effectively train spectral–temporal tokens and address the global pattern loss in spectral–visual feature fusion, we propose a frequency-domain fusion strategy (FFS), which includes two novel modules: the frequency attention network (FAN) and a Kolmogorov–Arnold network (KAN)-based convolutional unit (CuKAN). Inspired by the discrete cosine transform (DCT), which decomposes signals to different frequency components of the input feature, we introduce FAN. FAN decouples low-frequency and high-frequency information using self-adaptive weights to filter low-frequency features in the frequency domain. This allows the network to focus on global information derived from low-frequency features. To further prevent the local overfitting of features extracted by CNNs, we propose utilizing KAN to extract nonlinear features for frequency-domain attention computation, forming CuKAN. By leveraging parameterized spline functions and combination weights flexibly trained through convolutional layers, this method simulates complex relationships and achieves higher accuracy by optimizing parameters.

The main contributions of this article are as follows.

1) A spectral–temporal tracking Transformer based on a frequency-domain fusion strategy (S3T-FFS), termed S3T-FFS, is proposed. S3T-FFS allows for spectral–temporal feature modeling to capture the spectral invariant information in time series, enhancing the tracking performance in disturbed scenarios.
2) A spectral–temporal token is designed to associate the spectral information between the template and the search frames, which captures the spectral–temporal relationships of objects to form multiview representations, all without the online template updating.
3) An FAN is designed to guide the network to focus on global information by leveraging the DCT of decoupling frequency-domain representation. It avoids local overfitting in feature learning while obtaining robust fusion features.
4) A CuKAN is designed to further explore the nonlinear relationship of fused features and alleviate the local overfitting further. It uses learnable parameters to adjust the shape of the spline functions, effectively extracting nonlinear features. Extensive experiments on public datasets demonstrate the effectiveness of our proposed methods.

In this work, the visual modality refers to a three-channel false-color image, which is generated by selecting representative spectral bands from the HS cube. This false-color representation is not captured by a separate RGB sensor, but rather constructed directly from the HS data to simulate conventional appearance features and facilitate compatibility with RGB-pretrained tracking modules [8], [14]. The remaining article is organized as follows. Section II reviews the related works, including HS tracker, FANs, and Kolmogorov–Arnold network. Section III introduces the proposed S3T-FFS. Extensive ablation and comparative experimental results are shown in Section IV to show the effectiveness of the proposed methods on the HOTC-2020, HOTC-2024, and BihoT datasets. Finally, Section V concludes the article.

II. Related Work

A. HS Object Trackers

HS imaging captures rich spectral information across hundreds of contiguous and narrow bands, enabling fine-grained perception of material composition and structural properties.

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This unique capability has made HSI increasingly valuable in a wide range of applications, including remote sensing [15] and intelligent surveillance. In particular, tasks such as HS object detection, object tracking [16], and super-resolution reconstruction [17] benefit from the enhanced spectral discriminability of HSI data. For example, Li et al. [9] proposed a multiview fusion network to model spatial–spectral features. Li et al. [18] designed a loss function that allows denoising and classification to benefit each other during the learning process, significantly improving classification performance.

In contrast to conventional RGB tracking or detection, HOT leverages rich spectral information that extends beyond the visible bands. This enables more accurate discrimination of materials and surfaces, even when objects exhibit similar spatial or texture characteristics in RGB space. Jiang et al. [19] proposed PiV-AHPC, a novel 3-D HS object detection network. The method features a pillar-voxel dual-branch encoder that separately captures spectral and spatial cues, achieving strong generalization across diverse airborne scenarios. In the early stages of HOT research, researchers focused on manually extracting features, with most methods utilizing the correlation filtering to complete online feature learning and object-aware localization [20], [21], [22]. Uzkent et al. [23] proposed a kernelized correlation filter (KCF) to learn the location prediction of objects online. The features extracted by this method lack discriminability and robustness compared to the CNN features, resulting in poor performance. In recent years, the Siamese-based methods [7], [8], [24], [25] have received widespread attention, which extract features of the template and search images through CNN, and locate the object's position by designing a prediction network. Li et al. [8] proposed a Siamese network for band regrouping and feature extraction. Wang et al. [26] proposed a band regrouping method based on the Transformer and designed a feature refinement module to extract the discriminative features. Nowadays, most research focuses on combining spectral and visual information, improving the discriminability of features. The feature fusion-based methods [9], [14], [27] simultaneously utilize the object's spectrum and texture to form complementary feature representations. Wu et al. [28] proposed a bandwise spatial attention module to emphasize the salient area near the object of interest. Liu et al. [14] proposed a dual-stream network that simultaneously extracts the spectral and visual features by modifying the number of input channels in the ResNet [29].

B. Frequency Attention Networks

Recently, the combination of frequency-domain knowledge and attention mechanism has been proven to be an effective method of network design [30], [31], [32], [33], which utilizes frequency-domain analysis to enhance the feature extraction and optimizes the features through the attention mechanism further. Yun et al. [34] proposed a token mixer to capture high- and low-frequency components of input features, which improves the performance of models. Patro et al. [35] designed a novel visual backbone network called SpectFormer. The spectral and multihead attention layers are combined to refine the feature representation adaptively. Huang et al. [36] proposed an efficient tokenizer, which performs the frequency filtering by fast Fourier transform. With this global token mixer, the backbone network performs well in various downstream visual tasks. Ma et al. [37] built a learning-based frequency attention framework to enhance the interfeature representation. He et al. [38] proposed to modulate global illumination and local contrast through a wavelet-based attention network, which enhances the performance of models.

C. Kolmogorov–Arnold Networks in Computer Vision

Inspired by the Kolmogorov–Arnold representation theorem [39], KAN [40] was proposed to use learnable functions instead of fixed activation functions, fundamentally eliminating the dependence on linear weight matrices, which is completely different from the method of multilayer perceptrons (MLP). This design allows each weight parameter in KAN to be replaced by a univariate function, which is typically parameterized in the form of spline functions, providing high flexibility and the ability to simulate complex functions with fewer parameters. At present, the research on KAN in computer vision is still in its infancy. Cambrin et al. [41] proposed a KAN-based crop field segmentation framework and designed a novel gradient-based explanation method to prove the rationality of the proposed method. Wang et al. [42] proposed an image change detection framework named SpectralKAN to enhance the performance while requiring less calculation costs. Cheon [43] designed a satellite image classification method, named KCN, combining CNN and KAN, which proves the effectiveness of this hybrid structure. Li et al. [44] proposed a U-Net-like KAN, termed U-KAN, to improve backbones for medical image segmentation and generation. A novel tokenized with the special KAN layers is designed to tokenize the medical features.

The existing frequency-domain attention methods, such as FCA-Net, focus on enhancing global representation by applying Fourier or cosine transforms to spatial features, but do not consider spectral–temporal consistency or cross-modal fusion. In contrast, FFS introduces two task-specific modules—FAN and CuKAN—to address challenges unique to HS tracking, such as adjacent appendage interference and local feature overfitting. FAN adaptively filters frequency components to emphasize global patterns, while CuKAN leverages nonlinear spline functions to extract robust features in the frequency domain. This design enables more effective fusion of spectral and pseudo-visual modalities, which is not addressed by previous frequency-based attention frameworks.

III. Proposed Method

The proposed S3T-FFS includes a temporal Transformer tracking backbone, FAN, and CuKAN. The temporal Transformer tracking backbone simultaneously extracts the visual features of RGB images and spectral features of HSIs and captures the object regions with spectral invariance through the proposed spectral temporal token. In addition, to address the issue of global pattern loss in feature fusion methods, we propose a frequency-domain fusion strategy that includes FAN and CuKAN. FAN decouples the low-frequency and high-frequency components of features and dynamically adjusts the

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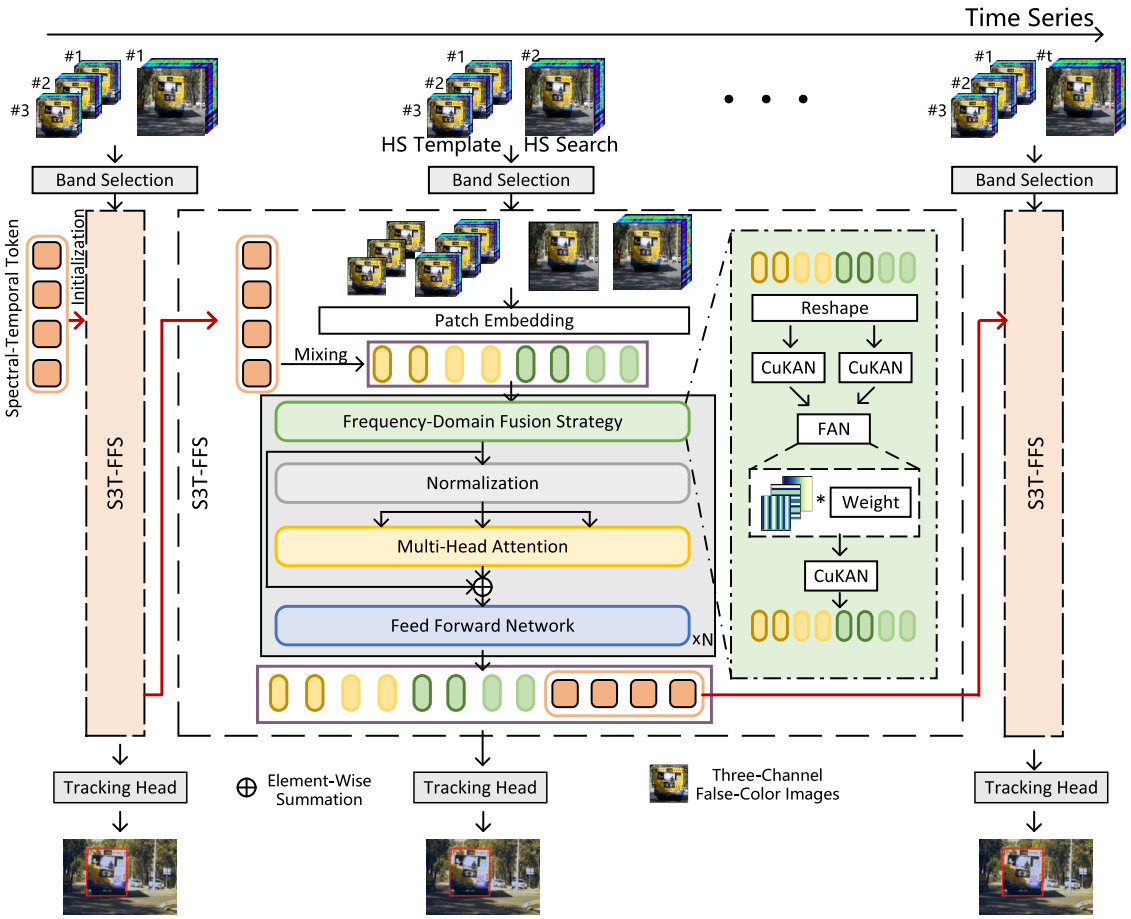


Fig. 2. Illustration of the overall structure of our proposed S3T-FFS, including the temporal Transformer tracking backbone and FFS. We construct a false-color modality by selecting three representative HS bands [14]. This false-color image is used to simulate RGB-like appearance features and is jointly utilized with the full HS cube in our multimodal fusion framework.

component weights to focus on global patterns. CuKAN alleviates local overfitting when using CNN to extract frequency features and collaborates with FAN to extract global features. After processing by FAN and CuKAN, the fusion features of the visual and spectral features can be obtained. We employ the existing tracking paradigms to model the relationship among temporal, template, and search tokens. Finally, the search token is input into the head network for result prediction.

A. Temporal Transformer Tracking Backbone

To address the issue of insufficient modeling of temporal information in the existing methods, we propose a temporal tracking backbone to mine spectral relationships in temporal data. Specifically, we change the traditional image-level sampling to video-level sampling, and the transformer tracker receives a series of consecutive HS frames as input. To utilize the complementary information of spectra and vision fully, we follow the advanced prompt tracking paradigm [45] and fine-tune our network training on a foundation model [46], which can handle the tracking problem of HS and visual images. Our visual input is generated by selecting three fixed bands from the HS image, including the 1st, 9th, and 15th bands following [14], to form a three-channel false-color image. The two templates from HS and visual modality are represented as $T_{\text{hs}} \in \mathbb{R}^{C_{\text{hs}} \times H_T \times W_T}$ and $T_{\text{vis}} \in \mathbb{R}^{3 \times H_T \times W_T}$, respectively. The two searches from HS and visual modality are represented as $S_{\text{hs}} \in \mathbb{R}^{C_{\text{hs}} \times H_S \times W_S}$ and $S_{\text{vis}} \in \mathbb{R}^{3 \times H_S \times W_S}$, respectively. Among them, C_{hs} represents the number of bands, H_T and H_S represent the heights of the template and search frames, and W_T and W_S represent the widths of the template and search frames, respectively. The Transformer tracker receives a series of image patches as input. Thus, a patch embedding layer is utilized to convert each 2-D channel into a 1-D token. Taking the template image as an example, it can be expressed as follows:

$$H_{\text{hs}}^T = Em\left(T_{\text{hs}}\right), H_{\text{hs}}^T \in \mathbb{R}^{D \times N_T} \tag{1}$$

$$H_{\text{vis}}^T = Em\left(T_{\text{vis}}\right), H_{\text{vis}}^T \in \mathbb{R}^{D \times N_T} \tag{2}$$

where $Em(\cdot)$ denotes the embedding layer, $D = 768$ denotes the embedding dimension, and N_T denotes the number of tokens. Considering that in a time series, for a given region of interest, the tracker should perceive regions with the same spatial and spectral features between the template and the search frames in a time window. We propose a spectral–temporal token $H_t \in \mathbb{R}^{1 \times D}$ to capture object regions with spectral invariance, which transfers the visual and spectral information of the chosen object in the subsequent frames. This token is initialized as a learnable parameter without predefinition. After that, a fusion token can be generated from the spectral and visual tokens using the FFS. Finally, the token sequences

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are concatenated with each other to represent input sequences $\{H_{\text{hs}}^T, H_{\text{vis}}^T\}$ and $\{H_{\text{hs}}^S, H_{\text{vis}}^S\}$ formally.

In this backbone network, we employ two information transfer strategies to model the relationship among the temporal, template, and search tokens, including the temporal-to-template strategy and the temporal-to-search strategy. To achieve this, the token sequences are first processed by the following proposed FFS, and the input sequence can be first transformed into $\left\{H_t, H_{\text{fuse}}^T, H_{\text{fuse}}^S\right\}$. The encoder of the Transformer completes the first strategy. After the multihead self-attention calculation of the Transformer, the spectral–temporal token can aggregate the information on all other tokens, achieving global feature aggregation in the timeline. It can be formulated as follows:

$$\left\{H_t^l, H_{\text{fuse}}^{T,l}, H_{\text{fuse}}^{S,l}\right\} = \text{MSA}_{l-1}\left(\left\{H_t^{l-1}, H_{\text{fuse}}^{T,l-1}, H_{\text{fuse}}^{S,l-1}\right\}\right) \tag{3}$$

where $l = 1, 2, \ldots, L$, MSA denotes the multihead self-attention calculation. In this way, the spectral information can be transferred between the spectral time-series trajectory and the template images. The second strategy aims to complete the information transfer between the spectral–temporal token and the search token. From Fig. 2, it can be seen that after the Transformer network extracts the features from the current search frame, the spectral–temporal token is propagated to extract the feature of the following search frames to update the historical frame information and guide the learning of the following frames. Within the two transmission strategies, the tracker captures the invariant spatial–spectral information and transfers it between different frames, improving the tracking performance.

B. Frequency-Domain Fusion Strategy

HSIs provide richer spectral information than traditional visual images and exhibit strong robustness against interference factors such as lighting changes and background clutter (BC). Therefore, learning an effective fusion feature representation is a critical challenge. However, existing modality fusion methods often suffer from global pattern loss, which can lead to localization errors. To address this issue, we propose the FFS, which includes two key components: FAN and CuKAN.

FAN: The 2-D DCT (2D-DCT) is a commonly used transformation technique in the image processing to transform images from the spatial domain to the frequency domain. It helps extract features and perform compression and denoising by decomposing the frequency components of the image. The 2-D DCT represents an image as a weighted sum of cosine waves of different frequencies, where the low-frequency components typically contain the majority of the image's essential information [47]. The function of 2-D DCT with basic function B can be typically formulated by the following equation:

$$f_{h,w}^{2d} = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} x_{i,j}^{2d} B_{h,w}^{i,j} \tag{4}$$

$$B_{h,w}^{i,j} = \cos\left(\frac{\pi h}{H}\left(i+\frac{1}{2}\right)\right)\cos\left(\frac{\pi w}{W}\left(j+\frac{1}{2}\right)\right) \tag{5}$$

where $h \in \{0, 1, \ldots, H-1\}$, and $w \in \{0, 1, \ldots, W-1\}$. H and W denote the height and width of 2-D images, respectively, and $f_{h,w}^{2d}$ denotes the frequency components. We omit some constant normalization factors. Considering that low-frequency features help capture global information and overall patterns of input, while high-frequency features tend to capture details and local variations of features, we propose a frequency attention mechanism based on the 2-D DCT to extract more low-frequency components, including a *Transformation and Selection* process.

Transformation: Given an input feature $x \in \mathbb{R}^{C\times H\times W}$, the transformed result of C-dimensional vectors can be obtained by calculating the corresponding 2-D DCT frequency components

$$\begin{aligned}\text{Feqc}^i(x) &= \text{2DDCT}^{m_i,n_i}(x)\\ &= \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} x_{:,h,w} B_{h,w}^{m_i,n_i}\end{aligned} \tag{6}$$

where $[m_i, n_i]$ represent the ith frequency component indices. Different from FCA-Net [48], FAN selects the frequency of complete channel information, avoiding the bias caused by feature partitioning.

Selection: A frequency component selection mechanism is proposed for the adaptive fusion of different frequency components. This selection process can be formulated by the following equation:

$$\text{Feqc}(x) = \sum_{i}^{N} w^i \text{Feqc}^i(x) \tag{7}$$

where N denotes the number of frequency components and w_i denotes the ith frequency weight. After the above two steps, we obtain a C-dimensional compressed vector that perceives the global pattern of features. Upon the compressed information, the frequency attention weights are obtained from a fully connected layer FC, and a Sigmoid activation function, which can be formulated by the following equation:

$$\text{FeqAtt}(x) = \text{Sigmoid}\left(\text{FC}\left(\text{Feqc}(x)\right)\right). \tag{8}$$

In the S3T-FFS network, FAN is proposed to extract the spatial–spectral global information of the template and search tokens, respectively. The network structure is shown in Fig. 3. The template tokens $\{H_{\text{hs}}^T, H_{\text{vis}}^T\}$ are first resized to $\left\{f_{\text{hs}}^T \in {}^{D\times\sqrt{N_T}\times\sqrt{N_T}}, f_{\text{vis}}^T \in {}^{D\times\sqrt{N_T}\times\sqrt{N_T}}\right\}$, which is represented by the following equation:

$$f_{\text{hs}}^T = \text{Resize}\left(H_{\text{hs}}^T\right) \tag{9}$$

$$f_{\text{vis}}^T = \text{Resize}\left(H_{\text{vis}}^T\right). \tag{10}$$

The HS feature f_{hs}^T is processed by the *Transformation and Selection* process, and after obtaining the attention weight, it is multiplied elementwise with the original feature. Finally, the residual connections are adopted to complete the cross-modality fusion with the visual features. This process can be expressed as follows:

$$f_{\text{fuse}}^T = \text{FeqAtt}\left(f_{\text{hs}}^T\right) \odot f_{\text{hs}}^T + f_{\text{vis}}^T \tag{11}$$

$$H_{\text{fuse}}^T = \text{Flatten}\left(f_{\text{fuse}}^T\right) \tag{12}$$

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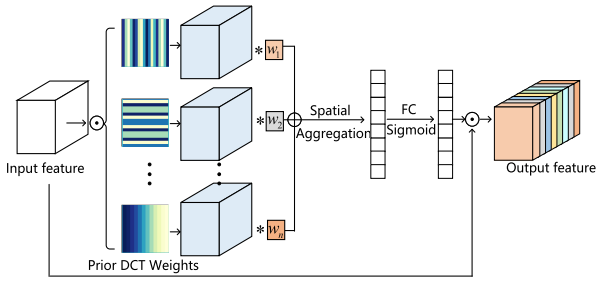


Fig. 3. Illustration of the proposed FAN. The input feature is convolved with a set of prior DCT-based frequency kernels to extract frequency-specific responses. The final output is obtained by reweighting the input feature, enabling the network to focus on discriminative frequency components.

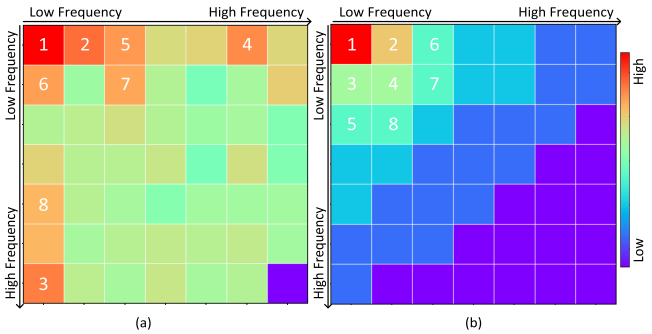


Fig. 4. Illustration of the confusion matrices of performance from the frequency domain. (a) Performance ranking. (b) Low frequency.

where $\odot$ denotes the elementwise multiplication. The fused token H^T_{fuse} can be obtained from the flattened f^T_{fuse}. For simplicity, we omit the introduction of the search tokens here, which is the same process as the template tokens.

It is worth noting that predefining the frequency components of 2-D DCT is a crucial step, as it determines the number of priority DCT weights in the FAN, which corresponds to the amount of decoupled low- and high-frequency information. To select the most effective frequency combinations and reduce computational complexity, we divide the frequency space of the DCT into 7×7 subspaces, with each part corresponding to a specific frequency-domain component. Additionally, prior knowledge is introduced to assist in setting these frequency combinations. Fig. 4 illustrates the confusion matrix in [48]. Fig. 4(a) shows the top 8 frequency components in terms of performance, while Fig. 4(b) highlights the top 8 low-frequency components. To investigate the impact of different frequency combinations on the experimental results, we conduct the ablation experiments in Section IV. Specifically, we select the top 4 frequency components of performance to generate the prior DCT weights in FAN.

Kolmogorov–Arnold Network-Based Convolutional Unit: Due to the integration of spline functions and learnable activation functions into its framework, KAN has been demonstrated to possess greater flexibility and nonlinear representation capabilities compared to the traditional CNN and MLP networks, which are based on the theorem of KANs. While CNNs are effective at capturing the local patterns and edge information in images, they tend to overemphasize local features or noise, which can limit the extraction of the low-frequency features in FAN, leading to local overfitting and reduced generalization performance. To address this issue, we introduce KAN into the convolutional structure and form the FFS in conjunction with the FAN to alleviate this issue.

1) KANs Revisit: The Kolmogorov–Arnold representation theorem points out that a multivariate continuous function f can be represented as a combination of a finite number of univariate continuous functions, which can be formulated by the following equation:

$$f(x_1, \ldots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^{n} \varphi_{q,p}(x_p) \right) \tag{13}$$

where $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $\varphi_{q,p} : [0,1] \to \mathbb{R}$ denote the univariate functions of each input variable x_p. n is the number of elements in the input.

However, these univariate continuous functions are nonsmooth and even fractal, so the Kolmogorov–Arnold representation theorem struggles to handle the deep learning task. With the analogy between KAN and MLP networks [40], we can see that the above theory is actually a two-layer KAN. Clearly, a single-layer KAN layer with n_{in} dimensional input and n_{out} dimensional output can be defined as a matrix of 1-D functions

$$\Phi = \{\varphi_{q,p}\} \tag{14}$$

where $p = 1, 2, \ldots, n_{\text{in}}$ and $q = 1, 2, \ldots, n_{\text{out}}$. The $\varphi_{q,p}$ are a series of trainable functions. $\varphi_{p,q}$ are learnable spline functions as follows:

$$\varphi_{p,q}(x) = \omega_a \cdot b(x) + \omega_b \cdot \text{spline}(x) \tag{15}$$

where ω_a and ω_b denote the trainable weight parameters, respectively, $b(\cdot)$ denotes the Sigmoid linear unit (SiLU) function, and spline$(\cdot)$ denotes a series of basis functions

$$\text{spline}(x) = \sum_i c_i B_i(x) \tag{16}$$

where $B_i(\cdot)$ is the B-spline function and c_i denotes optimized coefficients during training. We can observe that the inner function inside the summation of the right-hand side of (13) forms a KAN layer with $n_{\text{in}} = n$ and $n_{\text{out}} = 2n + 1$, while the outer function forms a KAN layer with $n_{\text{in}} = 2n + 1$ inputs and $n_{\text{out}} = 1$. Based on this, the activation function connecting the ith node in layer l-1 and the jth node in layer l can be formulated as follows:

$$\varphi_{l-1,j,i}, l = 1, \ldots, L, \; i = 1, \ldots, n_{l-1}, \; j = 1, \ldots, n_l. \tag{17}$$

The jth node of the lth layer can be represented as follows:

$$x_{l,j} = \sum_{i=1}^{n_{l-1}} \varphi_{l-1,j,i}(x_{l-1,i}), \quad j = 1, \ldots, n_l \tag{18}$$

where $x_{l,j}$ and $x_{l-1,i}$ denote the activation values of the jth node of the lth layer and the ith node of the l-1th layers, respectively. According to (13) and (18), it can be rewritten as follows:

$$x_l = \phi_{l-1}(x_{l-1}) = \begin{pmatrix} \varphi_{l-1,1,1}(\cdot) & \cdots & \varphi_{l-1,1,n_{l-1}}(\cdot) \\ \vdots & \ddots & \vdots \\ \varphi_{l-1,n_l,1}(\cdot) & \cdots & \varphi_{l-1,n_l,n_{l-1}}(\cdot) \end{pmatrix} x_{l-1} \tag{19}$$

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where $x_{l-1} \in {}^{n_{l-1}}$ denotes the lth layer input of KANs and $x_l \in {}^{n_l}$ denotes the output. Thus, the complete KAN structure can be expressed by the following formula, which adapts to the complex relationships in the data through the flexibility of the spline functions:

$$\mathrm{KAN}(x) = (\phi_{L-1} \circ \phi_{L-2} \circ \cdots \circ \phi_0)\, x \tag{20}$$

where $\circ$ denotes the operation of composite functions.

2) CuKAN: As shown in (13), the Kolmogorov–Arnold theorem restricts the number of hidden layers, limiting the number of nodes in each hidden layer to $2n + 1$ [42]. The KAN eliminates this limitation, allowing for the construction of networks with any dimension and number of layers to handle more complex problems. Based on this, in the proposed CuKAN, we combine CNN and KAN layers based on (14) and (15), using CNNs to learn the two weight parameters in (14). The network structure is shown in Fig. 5. Compared with the pure CNN structure, we replace the original convolutional kernel weights with a learnable continuous function φ. The coefficients of the activation function and the spline function are integrated. The modified kernel can be formulated as follows:

$$\mathrm{CuKAN}(x) = \mathrm{Cov}_a(b(x)) + \mathrm{Cov}_b(B_i(x)) \tag{21}$$

where Cov_a and Cov_b are two convolutional layers, respectively. In the network training, we replace the standard CNN kernel calculation method with (14) to form the CuKAN. This approach not only allows for the seamless updating of KAN parameters during training via gradient descent but also leverages the nonlinear modeling capabilities of KANs. By doing so, CuKAN helps mitigate the excessive focus on local features that CNNs often exhibit. Within the FFS framework, we first use CuKAN for global feature extraction before passing the features into the FAN module. After performing the frequency attention calculations and the feature fusion, we apply the CuKAN again for the nonlocal feature embedding, further enhancing the overall feature representation.

C. Loss Function

After extracting the features through the above backbone network, we employ a head network to obtain the classification response map $M_c \in [0,1]^{\frac{H_s}{P} \times \frac{W_s}{P}}$, size map $M_s \in [0,1]^{2 \times \frac{H_s}{P} \times \frac{W_s}{P}}$, and offset map $M_o \in [0,1]^{2 \times \frac{H_s}{P} \times \frac{W_s}{P}}$. P denotes a downsampling ratio. The prediction box can be obtained by the following equation:

$$\begin{aligned}(x, y, w, h) = &\left(i_p + M_o\left(0, i_p, j_p\right), j_p + M_o\left(1, i_p, j_p\right)\right.\\ &\left. M_s\left(0, i_p, j_p\right), M_s\left(1, i_p, j_p\right)\right).\end{aligned} \tag{22}$$

where $(i_p, j_p) = \arg\max_{(i,j)} M_c(i, j)$, which denotes the position of the object where the maximum value in M_c appears. The proposed S3T-FFS is trained with a multitask loss, which can be formulated by the following equation:

$$L = L_{\mathrm{cls}} + \lambda_1 L_{\mathrm{iou}} + \lambda_2 L_1 \tag{23}$$

where L_{cls} denotes the focal loss for classification [49], L_{iou} denotes the IoU loss [50], and L_1 loss is used for regression. λ_1 and λ_2 are set to 2 and 5, respectively, according to [46].

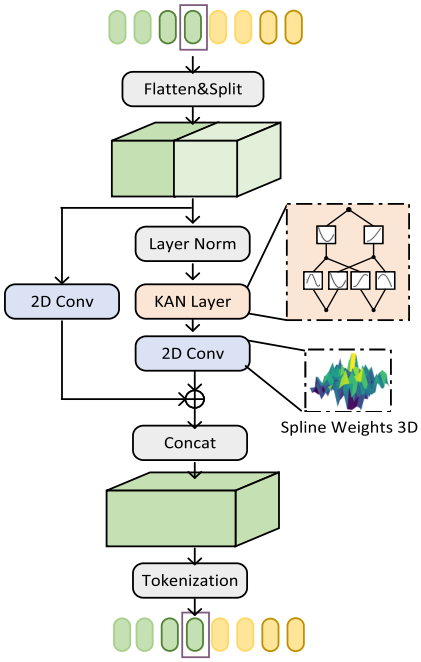


Fig. 5. Illustration of the proposed CuKAN. The input sequence is first flattened and split into local patches, followed by normalization and processing through a dual-branch structure. The extracted features are concatenated and tokenized to form the final output, enabling adaptive modeling of local spatial and semantic patterns.

IV. Experiments

A. Experimental Settings

Datasets: In the experiment, HOTC-2020, HOTC-2024, and BihoT datasets are used to evaluate the proposed S3T-FFS. The details are described as follows.

1) HOTC-2020 Dataset: HOTC-2020 was constructed at the 2020 Hyperspectral Object Tracking Challenge. It contains 75 HS videos of 16 bands, including 12 839 frames for training and 16 574 frames for testing. The dataset contains 11 challenging factors, including BC, DEF, fast motion (FM), illumination variation (IV), in-plane rotation (IPR), low resolution (LR), motion blur (MB), OCC, out-of-plane rotation (OPR), out-of-view (OV), and scale variation (SV).

2) HOTC-2024 Dataset: HOTC-2024 was constructed at the 2024 Hyperspectral Object Tracking Challenge. It includes three types of band numbers: 15 bands, 16 bands, and 25 bands, corresponding to RedNIR, VIS, and NIR, respectively. Each video frame is stored as a 2-D image, which can be processed into a 3-D image cube using publicly available algorithms. The complete training set contains 218 HS videos, while the validation set contains 118 HS videos, each with bounding box annotations for the tracked object in every frame. The challenging factors are the same as HOTC-2020.

3) BihoT Dataset: [51]. BihoT is the first dataset used for HS camouflage object tracking, with 41 912 images captured by an IMEC XIMEA MQ022HG-IM-SM5X5-NIR camera in the 25 bands across the wavelength range of 600–974 nm, including 22 992 images in the training set and 18 920 images in the testing set. This range partially covers the visible spectrum. Here, spectral information refers to the data captured within the dataset's defined wavelength interval. The dataset

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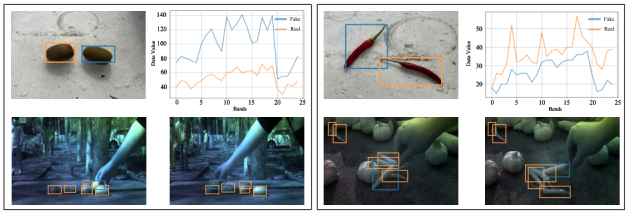


Fig. 6. Examples of spectral reflectance curves for camouflaged objects, i.e., real objects and their plastic replicas, from the BihoT dataset. Despite strong visual similarity in false-color images, the materials exhibit significantly different spectral characteristics across wavelengths. This highlights the critical role of HS information in distinguishing objects from confusing backgrounds, especially under camouflage conditions.

is annotated with nine attributes, including BC, FM, IPR, IV, LR, OCC, OPR, spectral consistency (SC), and spectral variation (SV). Unlike typical tracking datasets, BihoT focuses on challenging scenes where objects are visually similar in RGB space but can be distinguished via spectral information. Some representative samples are shown in Fig. 6.

a) Evaluation Metrics: We use mainstream success and precision plots to measure the effectiveness of the proposed algorithm. All trackers follow the rule of one-pass evaluations. The success plots measure the proportion of annotated boxes and predicted IoUs that are less than the threshold, while the precision plots refer to the proportion of pixel distances between the center points of annotated boxes and predicted boxes that are less than the threshold. In the specific comparison, we use the area under the curve of success plots (AUC) and the precision plots with a threshold of 20 pixels (DP_20) to compare the tracking performance.

b) Implementation Details: All experiments are conducted using an NVIDIA RTX 4090 GPU. The backbone is first pretrained on the COCO [52], ImageNet VID [53], and YouTube-BB [54] datasets. The template and the search patches are set to 384×384 pixels and 192×192 pixels, respectively. An AdamW optimizer with a weight decay of 0.0001 is utilized to optimize the parameters of the network during training. The batch size is set to 8, and a total of 10 epochs is adopted.

B. Comparison With State-of-the-Arts

1) Comparison on the HOTC-2020 Dataset: To demonstrate the superiority of our proposed method, we first conduct comparative experiments with state-of-the-art HS trackers. The comparison methods can be broadly classified into two categories: feature fusion and false-color image inference. The first category includes methods such as SiamHYPER [14], MMF-Net [9], and SST-Net [27]. Specifically, SiamHYPER proposes a spatial–spectral fusion method based on the channel attention mechanism. MMF-Net proposes a multiview tracking framework that incorporates material, spatial, and spectral information. SST-Net leverages the temporal–spatial attention mechanism to capture the discriminative cues of tracking objects. The second category, feature fusion-based methods, includes SENSE [25], SPIRIT [24], SEE-Net [8], and SpectralTrack [13]. SENSE proposes a spectral–spatial self-expression module to obtain false modalities. SPIRIT designs a spectral awareness interaction network to regroup the HSIs for false-color image inference. SEE-Net utilizes a self-encoder-based band reorganization method that converts HSIs into multiple three-channel false-color images for position prediction. SpectralTrack is a temporal-based HS tracking framework that integrates spectral–spatial fusion and memory enhancement. A memory perception module is used to inject temporal prompts, improving temporal reasoning.

The experimental results are listed in Table I. The AUC score of S3T-FFS reaches 0.691, and DP_20 achieves 0.959, highlighting the superiority of our method. The relatively poor performance of false-color image inference methods compared to feature fusion-based methods is primarily due to the loss of spectral information when converting HS data into three-channel images. Although some methods attempt to mitigate this issue by weighting inference results, these methods struggle in more challenging scenarios, such as camouflaged object tracking.

In contrast, our method considers both temporal and spectral information, effectively capturing spectral invariant regions across consecutive video frames, providing a significant advantage over false-color-based methods. Notably, our method outperforms the best-performing false-color-based temporal tracker, SpectralTrack, by 0.954 in DP_20. Furthermore, S3T-FFS demonstrates superior performance compared to feature fusion-based methods. For instance, MMF-Net, which considers spatial, spectral, and material information, achieves an AUC score of 0.691. However, our method surpasses MMF-Net by 2.7% in DP_20, indicating the effectiveness of our utilization of temporal information.

To further demonstrate the effectiveness of our proposed method, we conduct comparison experiments on the HOTC-2020 dataset with state-of-the-art RGB trackers, including ARTrack [59], MixFormer [60], OSTrack [46], SiamCAR [57], SiamGAT [56], SimTrack [58], and TransT [55]. The experimental results are listed in Table I, where all RGB trackers are tested on the false-color images provided in the HOTC-2020 dataset. The results clearly show that our method outperforms the SOTA RGB trackers by a significant margin. Additionally, it is evident that in most cases, RGB trackers struggle to perform as well as HS trackers that utilize both spatial and spectral information. This highlights the importance of leveraging spectral information to handle BC and camouflaged objects effectively.

2) Comparison on the HOTC-2024 Dataset: To demonstrate the generalizability of our method and its ability to handle different numbers of bands, we conduct comparative experiments on the NIR and RedNIR datasets of HOTC-2024, denoted as HOTC24-NIR and HOTC24-RedNIR. Compared to HOTC-2020, the NIR dataset consists of 25 bands of HSIs, while the RedNIR dataset consists of 15 bands. The experimental results are listed in Table II, which indicate that our proposed method continues to outperform state-of-the-art HS and RGB trackers.

As for HOTC24-NIR, the AUC score reaches 0.790, which is 4.0% higher than the best-performing HS tracker and 4.2% higher than the best-performing RGB tracker. Similarly, the DP_20 score reaches 0.973, outperforming the best HS tracker

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TABLE I

OVERALL PERFORMANCE COMPARISON OF HS AND VISUAL TRACKERS ON THE HOTC-2020 DATASET. THE HS TRACKERS DEAL WITH THE HSIS, AND THE VISUAL TRACKER RUNS ON THE FALSE-COLOR IMAGES. UFB [25] AND FB DENOTE THE ATTEMPT TO USE FULL BANDS AND FEATURE FUSION, RESPECTIVELY. WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

Tracker	Years	Backbone	UFB	FB	AUC	DP_20
TransT [55]	2021	ResNet-50	No	No	0.633	0.879
SiamGAT [56]	2021	ResNet-50	No	No	0.581	0.827
SiamCAR [57]	2022	ResNet-50	No	No	0.586	0.846
SimTrack [58]	2022	ViT	No	No	0.600	0.845
OSTrack [46]	2022	ViT	No	No	0.557	0.816
ARTrack [59]	2024	ViT	No	No	0.689	0.935
Mixformer [60]	2024	ViT	No	No	0.677	0.934
SST-Net [27]	2021	VITAL	Yes	Yes	0.623	0.917
SiamHYPER [14]	2022	SiamFC	Yes	Yes	0.678	0.945
SEE-Net [8]	2023	SiamFC	Yes	No	0.654	0.907
SPIRIT [24]	2024	SiamFC	Yes	Yes	0.679	0.925
MMF-Net [9]	2024	SiamFC	Yes	Yes	0.691	0.932
SENSE [25]	2025	SiamFC	Yes	No	0.689	0.951
SpectralTrack [13]	2025	ViT	Yes	Yes	0.727	0.954
S3T-FFS	2025	ViT	Yes	Yes	0.691	0.959

TABLE II

OVERALL PERFORMANCE COMPARISON OF HS TRACKERS ON THE HOTC-2024 DATASET. WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

Band Num.	Metrics	SPIRIT [24]	SENSE [25]	SiamBAN [61]	ODTrack [62]	ARTrack [59]	SpectralTrack [13]	S3T-FFS (Ours)
15	AUC	0.379	0.367	0.346	0.492	0.423	0.539	0.565
	DP_20	0.516	0.465	0.456	0.615	0.540	0.705	0.735
25	AUC	0.655	0.561	0.547	0.748	0.640	0.750	0.790
	DP_20	0.823	0.765	0.741	0.927	0.785	0.937	0.973

TABLE III

OVERALL PERFORMANCE COMPARISON OF HS AND RGB TRACKERS ON THE BIHOT DATASET. THE HS TRACKERS DEAL WITH THE HSIS, AND THE VISUAL TRACKER RUNS ON THE FALSE-COLOR IMAGES. WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

Trackers	ODTrack [62]	SiamBAN [61]	ARTrack [59]	MixFormer [60]	SEE-Net [8]	SiamHYPER [14]	S3T-FFS (Ours)
AUC	0.406	0.343	0.417	0.388	0.315	0.403	0.708
DP_20	0.485	0.445	0.487	0.454	0.422	0.539	0.847

TABLE IV

COMPUTATION COMPLEXITY AND RUNNING TIME COMPARISON RESULTS INCLUDING FLOPS, MODEL SIZE, AND FPS. THE RGB TRACKERS RAN ON THE THREE-CHANNEL FALSE-COLOR IMAGES, AND THE HS TRACKER RAN ON THE HSIS

Methods	ODTrack [62]	SiamBAN [61]	ARTrack [59]	MixFormer [60]	SEE-Net [8]	SiamHYPER [14]	S3T-FFS (Ours)
FLOPs	32.55G	59.59G	37.15G	26.11G	297.99G	119.97G	34.91G
Size	92.12M	53.93M	173.10M	58.55M	53.93M	107.90M	97.27M
FPS	23.6	15.8	20.1	30.9	12.0	12.6	22.6

by 3.6% and the best RGB tracker by 4.6%. These results clearly demonstrate the robustness and superiority of our method across different spectral configurations. It is worth noting that our method achieves better performance over the temporal-based method (SpectralTrack), demonstrating the effectiveness of our method of incorporating temporal cues into HS video tracking.

3) Comparison on the BihoT Dataset: We also conduct comparative experiments on the camouflaged object tracking dataset BihoT, which is introduced to address BC and similar object interference. The experimental results are listed in Table III, which demonstrate that in scenarios where visually similar objects lack the distinctive texture features, the performance of RGB trackers relying on texture information and HS trackers using false-color image inference deteriorates. The best-performing tracker (ARTrack) achieves an AUC score of only 0.417 and a DP_20 of 0.487. In contrast, our method, which employs a more effective prompt-learning-based modal fusion strategy, achieves a higher AUC score of 0.708 and a DP_20 of 0.847, outperforming the compared trackers. This improvement highlights the robustness of our method in handling camouflaged objects, even in the absence of visually discriminative properties.

Besides, to evaluate the efficiency of the proposed tracker, we compare the computational complexity (FLOPs), model size, and tracking speed (FPS) with several representative RGB-based and HS tracking methods, as shown in Table IV. Our method achieves a good balance between computational efficiency and tracking speed. S3T-FFS requires only 34.91G FLOPs and 97.27M parameters, which is lower than most HS

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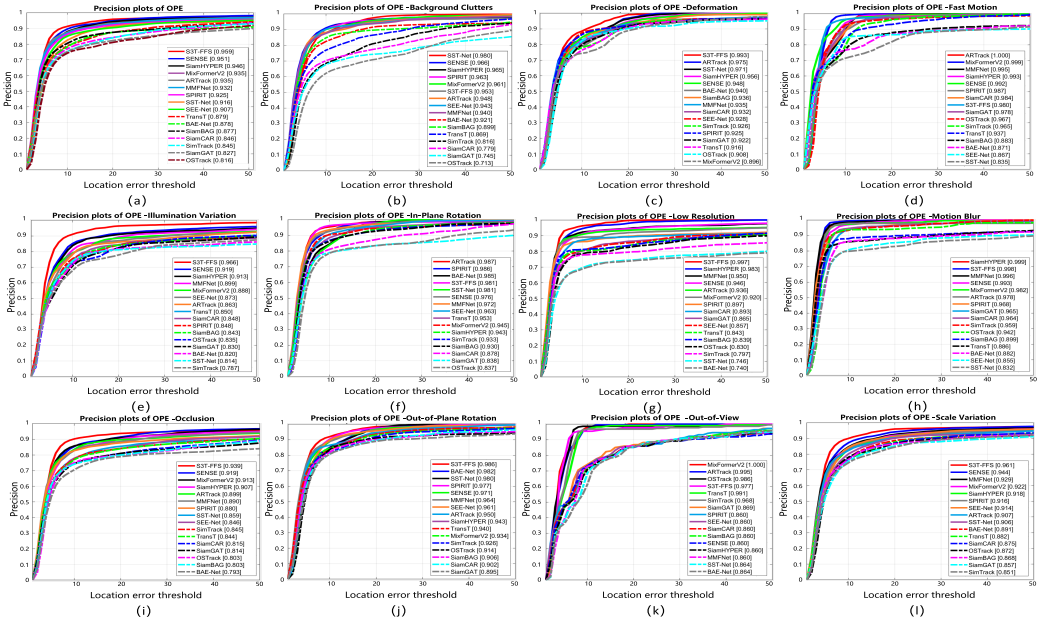


Fig. 7. Attribute-based comparison of precision plots. (a) Precision plot of OPE. (b) Precision plot of BCs. (c) Precision plot of DEF. (d) Precision plot of FM. (e) Precision plot of illustration variation (IV). (f) Precision plot of IPR. (g) Precision plot of LR. (h) Precision plot of MB. (i) Precision plot of OCC. (j) Precision plot of out-of-plane clutters (OPR). (k) Precision plot of OV. (l) Precision plot of SV.

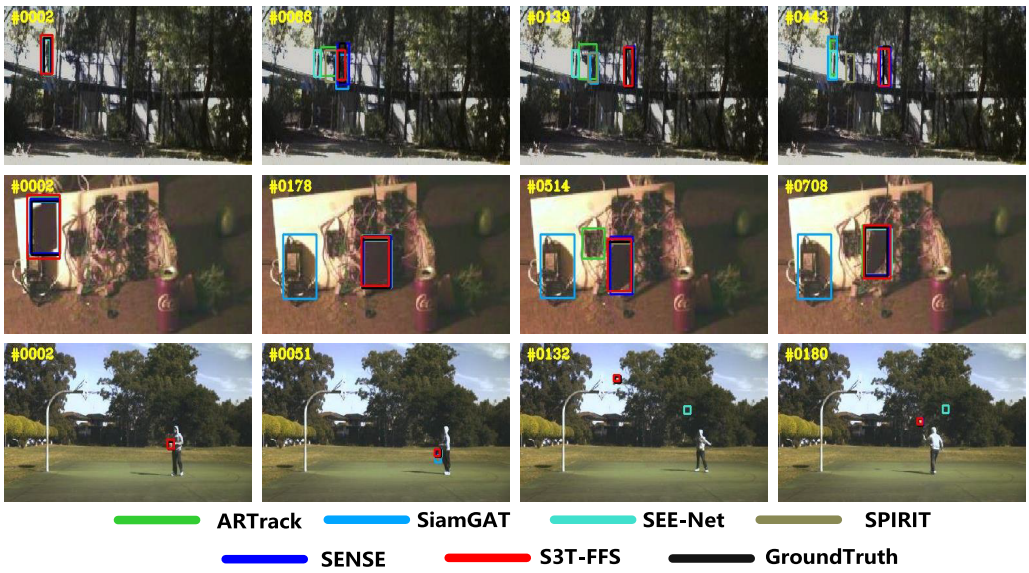


Fig. 8. Tracking results on the false-color images of campus, board, and basketball scenarios.

trackers. Moreover, it runs at 22.6 FPS, demonstrating real-time capability and outperforming the other HS-based methods in speed while maintaining a competitive model size.

C. Attribute-Based Evaluation

To further demonstrate the effectiveness of our method, we conduct an attribute-based comparison on the HOTC-2020 dataset, covering 11 challenging scenarios. The experimental results for the DP_20 metric are presented in Fig. 7. The proposed method achieves the best performance in 6 of 11 challenging scenarios. Notably, we attain the top results in the DEF, IV, and LR scenes, proving the ability of our method to effectively handle tracking problems where the object undergoes significant changes in the visual features. The superior performance of S3T-FFS is attributed to its ability to extract and leverage the spectral information of the object, which enhances its robustness to texture changes and visual disturbances.

D. Visualization Analysis

1) Tracking Results: We select three video clips (basketball, board, and campus) from the HOTC-2020 dataset and compare our method with state-of-the-art HS trackers. The results are presented in Fig. 8, where the ground-truth bounding box is framed by black lines, and the results of S3T-FFS are marked by red lines. The first column shows the visualization of the tracking box for the second frame, while the second to fourth columns illustrate the detection results for selected subsequent frames.

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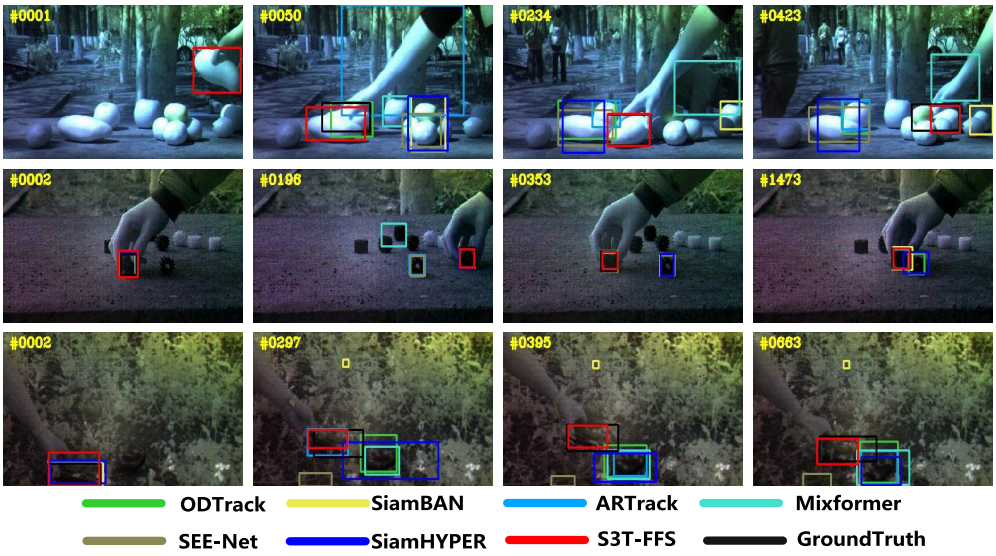


Fig. 9. Tracking results on the false-color images of mango, gear, and tank scenarios.

a) Campus Scene: In this scene, characterized by poor lighting conditions and significant BC, most trackers show a severe offset in the detection frames for subsequent frames, with only the initialization results in the first column being accurate. However, S3T-FFS is able to accurately predict the position of the detection box despite the challenging environmental conditions, demonstrating the robustness of the proposed algorithm. These results highlight the robustness and effectiveness of S3T-FFS in a variety of challenging tracking scenarios, particularly in environments with poor lighting, cluttered backgrounds, and rapid motion.

b) Board Scene: With a cluttered background and interference from visually similar objects, S3T-FFS continues to perform well, showcasing its ability to handle noise and distinguish objects in complex environments.

c) Basketball Scene: In this dynamic tracking scenario, S3T-FFS effectively captures the spectral invariance of the region of interest across consecutive frames, maintaining the superior tracking performance even under rapid motion.

Furthermore, we conduct additional visualization experiments on the BihoT dataset, which is specifically designed to evaluate tracking performance under camouflaged scenarios. We select three representative video sequences on the BihoT dataset for statistical visualization. These sequences are deliberately chosen to reflect the most challenging scenarios where overfitting is likely to occur, including the following: object-background confusion caused by camouflage (Tank sequence), interobject confusion where multiple similar objects coexist (Lemon sequence), and highly homogeneous backgrounds with minimal spatial contrast (Paper Sequence). As shown in the visual comparisons in Fig. 9, our method demonstrates superior robustness and accuracy in distinguishing camouflaged objects from visually similar objects. While the baseline methods often drift toward background distractors with overlapping shape, color, or motion cues, our method maintains consistent focus on the true object across frames. This is attributed to the frequency-aware fusion mechanism, which effectively captures global spectral–temporal patterns

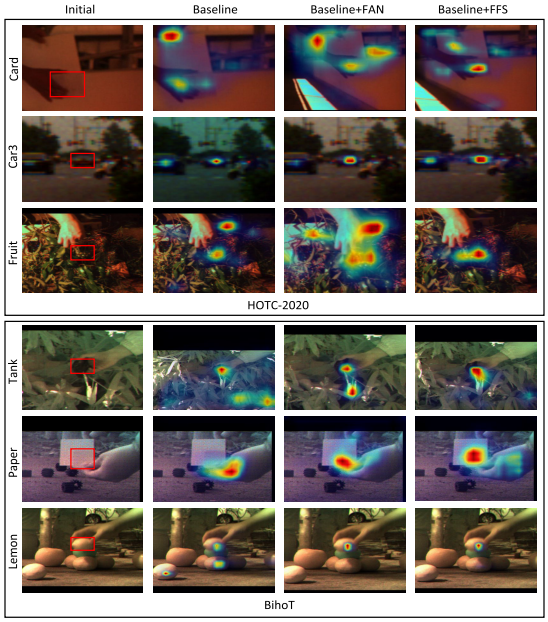


Fig. 10. Visualization of heatmaps on the HOTC-2020 and BihoT datasets. The red color represents a high response, while the blue color represents a low response.

and suppresses over-reliance on fine-grained local details. The results highlight the strength of our model in leveraging HS cues to resolve tracking ambiguities that are difficult to handle with conventional RGB-based or spatial-only methods.

Although our method achieves competitive performance on various benchmarks, we also observe certain failure cases under challenging conditions. As shown in Fig. 11, one typical failure occurs when there is a large visual inconsistency between the template and the object, such as when the template captures a horizontally oriented object while the object moves vertically in later frames. This discrepancy leads to tracking misalignment due to significant shape and appearance changes. Another common failure scenario involves severe OCC, where the object is largely blocked by surrounding objects for an

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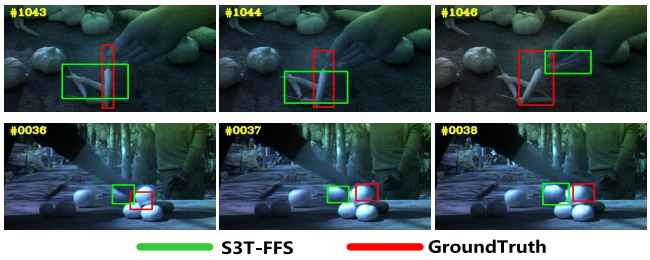


Fig. 11. Visualization of the failed tracking results on two video sequences from the BihoT dataset.

TABLE V
ABLATION EXPERIMENTAL RESULTS WITH THE BASELINE MODEL ON THE HOTC24-NIR DATASET. Δ INDICATES THE AMOUNT OF VARIATION

Methods	AUC	DP_20	Δ(AUC)	Δ(DP_20)
Baseline	0.759	0.932	-	-
+FCA-Net	0.773	0.952	+0.014	+0.020
+FAN	0.787	0.962	+0.028	+0.030
+CuKAN	0.790	0.973	+0.031	+0.041

extended period. In such cases, the model struggles to maintain reliable object representation, resulting in drift or complete object loss. These cases highlight the need for more robust motion modeling and OCC handling in future work.

2) Heatmap Visualization: To intuitively demonstrate the reduction of local overfitting in the original framework and the ability of the proposed FFS to extract global patterns, we utilize Grad-CAM++ [63] to visualize and analyze the feature heatmaps. We select several sequences with accessory interference from the HOTC-2020 dataset, including the card, car3, and fruit sequences. The experimental results are shown in Fig. 10, where brighter colors represent regions with a higher contribution to the network's performance. The first column shows the original image. The second column presents results from our implementation of the temporal Transformer tracking backbone based on visual features. The third column shows the results after adding FAN to the model. The fourth column displays the complete results of S3T-FFS. For the card sequence, the baseline method is influenced by things such as paper or hands, causing a shift in the response map. FAN helps the network focus on both low- and high-frequency information, allowing it to extract global patterns and avoid such interference, partially alleviating the issue. After introducing CuKAN—denoted as "Baseline+FFS"—the network's nonlocal feature extraction capability is further enhanced, enabling more accurate responses to the object's position, as shown in the fourth column. In more challenging scenarios, such as the car3 and fruit sequences, where visually similar objects appear, and severe OCC occurs, the baseline and FAN alone struggle to form effective feature representations and fail to locate the object correctly. However, the proposed FFS is able to extract more global patterns, leading to more accurate tracking results.

E. Ablation Experiments

1) Effectiveness of Each Component: We conduct ablation experiments on the HOTC24-NIR and BihoT datasets to demonstrate the effectiveness of the proposed components. The experimental results are listed in Table V. We first construct the spectral–temporal backbone based on [45] as the baseline model. The AUC score reaches 0.759, and the DP_20 reaches 0.932. Subsequently, to address the shortcomings of [48], a frequency component selection mechanism and a weighting strategy are proposed for our baseline model. Compared with the baseline model, the AUC score increases to 0.787, and DP_20 increases to 0.962. We further introduce a KAN-based convolutional unit for extracting global and nonlinear features, which yields an AUC score of 0.790 and DP_20 of 0.973. To show the effectiveness of FFS, we also employ the original frequency attention mechanism of FCA-Net based on the baseline model. It can be seen that the proposed method has improved by 0.014 and 0.010 in two indicators, respectively, compared with the traditional FCA-Net. For the BihoT dataset, as shown in Table VI, both the FAN and CuKAN modules contribute to the performance improvement on the BihoT dataset. Introducing FAN alone improves the AUC from 0.666 to 0.685 and DP_20 from 0.808 to 0.828. When CuKAN is further integrated, the performance increases to 0.708 in AUC and 0.847 in DP_20, achieving overall gains of +0.042 and +0.039, respectively, demonstrating the effectiveness of the proposed FAN and CuKAN modules for global spatial–spectral modeling.

TABLE VI
ABLATION EXPERIMENTAL RESULTS WITH THE BASELINE MODEL ON THE BIHOT DATASET. Δ INDICATES THE AMOUNT OF VARIATION

Methods	AUC	DP_20	Δ(AUC)	Δ(DP_20)
Baseline	0.666	0.808	-	-
+FCA-Net	0.671	0.811	+0.005	+0.003
+FAN	0.685	0.828	+0.019	+0.020
+CuKAN	0.708	0.847	+0.042	+0.039

TABLE VII
ABLATION EXPERIMENTAL RESULTS WITH DIFFERENT FREQUENCY COMPONENTS. Δ INDICATES THE AMOUNT OF VARIATION COMPARED WITH THE RESULTS OF THE FIRST LINE

Methods	AUC	DP_20	Δ(AUC)	Δ(DP_20)
top2	0.742	0.922	-	-
top4	0.787	0.962	+0.045	+0.040
top8	0.749	0.924	+0.005	+0.002
low2	0.741	0.917	-0.001	-0.005
low4	0.757	0.936	+0.015	+0.014
low8	0.771	0.961	+0.029	+0.039

2) Evaluation of the Number of Frequency Components: To assess the impact of varying frequency components, we conduct experiments with the FAN. Considering that different frequency components contain different information, we predefine the selection of frequency components according to the performance order and frequency order. First, regarding the performance order, we refer to the performance ranking in [48] and conduct experiments using the top-k performance ranking settings, which are denoted as "top2, top4, top8". In addition, we focused on exploring the influence of low-frequency components on experimental results regarding frequency order, which is denoted as "low2, low4, low8". The experimental results are listed in Table VII. We observe that the best experimental results are achieved with the top4 setting, where the AUC reaches 0.787, and DP_20 reaches 0.962. This setting shows significant advantages compared to settings that only utilize low-frequency components. These results demonstrate that considering both low-frequency and high-frequency

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TABLE VIII
ABLATION EXPERIMENTAL RESULTS OF DIFFERENT EMBEDDING DIMENSIONS OF FAN. Δ INDICATES THE AMOUNT OF VARIATION COMPARED WITH THE RESULTS OF THE FIRST LINE

Dimensions	AUC	DP_20	Δ(AUC)	Δ(DP_20)
8	0.787	0.962	-	-
16	0.758	0.935	-0.029	-0.027
32	0.769	0.946	-0.018	-0.016
64	0.754	0.938	-0.033	-0.024
128	0.758	0.939	-0.029	-0.023

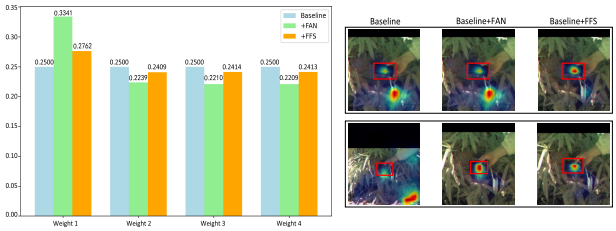


Fig. 12. Visualization of the learned frequency weight distributions and corresponding feature response maps on the BihoT dataset.

information enables the extraction of more discriminative features, ultimately improving tracking performance.

3) Evaluation of the Embedding Dimensions of FAN: The embedding dimension determines the amount of information used by flattened tokens for computing attention mechanisms in FAN. To investigate the impact of FAN embedding dimensions, we select multiple embedding dimension settings, including "8, 16, 32, 64, 128". The evaluated results are listed in Table VIII. It can be seen that the setting of dimension 8 achieves better experimental performance, with AUC reaching 0.787 and DP_20 reaching 0.962. Compared to other settings, this setting maintains better performance while reducing computational complexity.

4) Impact of Frequency Weight Distribution Changes in FFS on Feature Responses: To better understand the internal mechanisms of FAN and FFS, we visualize the learned frequency weight distributions and corresponding feature response maps in Fig. 12. The baseline model assigns uniform weights across all frequency components, leading to equal treatment of low- and high-frequency features. In contrast, FAN introduces adaptive weighting, dynamically increasing the contribution of informative frequencies (e.g., Weight 1 rises from 0.2500 to 0.3341), which enhances the model's ability to focus on salient foreground information. This is reflected in the heatmaps, where the object regions receive stronger and more concentrated responses. Furthermore, after incorporating CuKAN into the FFS module, the frequency weight distribution becomes smoother and more balanced, while still following the trend introduced by FAN. The resulting heatmaps show more consistent and precise attention on the object region, with reduced background noise. These observations demonstrate that FAN enhances frequency selectivity, while CuKAN further stabilizes the response patterns, jointly improving the model's robustness and resistance to overfitting.

V. CONCLUSION

In this article, we propose an S3T-FFS for HOT. First, we designed a spectral–temporal tracking backbone that simultaneously captures spatial, spectral, and temporal information using a novel spectral–temporal token. Then, we introduced an FFS, which includes two key components: the FAN and a CuKAN. The FAN decouples high-frequency and low-frequency information, adaptively adjusting the balance between these components to avoid local overfitting while focusing on more global information. The CuKAN uses learnable parameters to adjust the shape of the spline functions, fitting the training data more effectively and extracting nonlinear features, further mitigating local overfitting. Experimental results on the HOTC-2020, HOTC-2024, and BihoT datasets demonstrate the effectiveness of our proposed methods and their superiority over state-of-the-art methods. Besides, our method depends on RGB pretrained backbones, which may limit the generalization capacity in spectral domains that deviate significantly from natural RGB distributions. This may also hinder performance in cases where false-color bands are poorly correlated with RGB semantics. A promising direction for future work is to design dedicated hyperspectral-specific network architectures, which can better capture the rich spectral–spatial characteristics unique to HS video.

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